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MIBE: Multi-subject Interaction Benchmark and Evaluator for Personalized Image Generation

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 Revisions (/revisions?id=ia7gmI4SOA)

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TL;DR: We introduce MIBE, a unified benchmark and lightweight evaluator that overcomes traditional metric collapse in multi-subject generation to achieve strong human alignment.

Abstract:

Multi-subject personalized image generation requires the precise rendering of all requested reference identities and their specified interactions based on a guiding prompt. However, state-of-the-art models still struggle with this process, frequently omitting subjects, failing to preserve reference appearances, or misattributing interactions. Furthermore, existing metrics designed primarily for single-subject fidelity cannot reliably capture these errors, suffering severe degradation in ranking separability and failing to align with human preference as the subject count increases. To address this gap, we introduce Multi-subject Interaction Benchmark and Evaluator (MIBE), a unified framework comprising a Multi-subject Interaction Benchmark (MIB) and a Multi-subject Interaction Evaluator (MIE). MIB systematically covers diverse relation types and scene complexities through a decoupled data regime. This consists of a 60K-pair VLM-labeled Silver Set for scalable metric training and a 4K-pair double-blind Human Evaluation Gold Set covering a diverse range of state-of-the-art generators, with the Silver Set reaching 95.1% cross-VLM preference agreement. To demonstrate the utility of this benchmark, we present MIE, a lightweight, reference-conditioned evaluator trained exclusively on the Silver Set with a dual-head ranking and diagnosis objective. MIE exhibits strong cross-generator generalization on the Gold Set, achieving 0.922 overall pairwise accuracy against human preference, including 0.982 on seen generators and 0.884 on unseen generators. By outperforming a broad spectrum of baseline metrics, including CLIP and DINO variants, MIE demonstrates that diagnostic supervision can preserve ranking separability and human alignment where traditional evaluators collapse. Ultimately, by providing data with the diagnostic and preference labels required to train robust evaluators, MIBE serves as a comprehensive framework to benchmark multi-subject image generation and guide future model alignment.

Checklist Confirmation:  I confirm that I have included a paper checklist in the paper PDF.

Supplementary Material:  zip (/attachment?id=ia7gmI4SOA&name=supplementary_material)

Responsible Reviewing:  We acknowledge the responsible reviewing obligations as authors.

Primary Area: Computer vision (e.g., object detection, segmentation, image classification, image generation, video)

Secondary Area: AI for creatives (e.g., visual arts, music)

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Submission Number: 29259

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Summary:

This work presents an evaluation dataset and model aimed at addressing weaknesses in multi-subject generation with specified interactions. A dataset of fixed references and structured prompts are created in a hierarchical fashion to carefully evaluate performance in terms of existence (presence) of subjects, appearance of subjects, and following of interaction prompts. This dataset is split into a VLM-annotated and human annotated set. After performing training on the VLM annotated set, MIE is able to get high agreement with human evaluations and is much better suited for this evaluation than existing metrics.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

This work provides a very meticulous design of benchmark and evaluation addressing a significant gap in generative evaluation in the multi-subject multi-interaction setting. The extensively documented dataset generation process is followed up by thoughtful and targeted evaluation. Limitations are addressed sufficiently in section 5 in an honest and extensive fashion. This work is very strong and is very well-polished. The only "weaknesses" I observed are very minor recommendations. First of all, the actual generative model results are not presented so far as I could see, so it isn't immediately obvious how much room for growth there is on this task (although it seems this is significant). The number of examples prompts presented in the paper explicitly is very limited, although they are present in the appendix. The number of reference subjects is also fairly limited, given that identity-preservation is a key element the benchmark. It might be useful to see how well identity preserving extends for more obscure, long-tailed classes. Regardless, this work stands out in terms of all of the four desired elements: quality (very meticulous design and thorough experimentation), clarity (excellent documentation), significance (very practical benchmark and evaluator), and originality (devising an evaluation scheme in terms of pair-wise comparison as well as splitting the task into existence, appearance, interaction is a very interesting paradigm).

Quality: 4: excellent

Clarity: 4: excellent

Significance: 4: excellent

Originality: 4: excellent

Questions:

1. What were the scores for the models evaluated for the benchmark and which aspects of existence, appearance, and interaction do the current models fall short on?
2. What are the classes present in MIB and does MIE extend to unseen classes?
3. This seems like a great benchmark to maintain a leaderboard for!

Limitations:

Yes.

Rating: 6: Strong Accept: Technically flawless paper with groundbreaking impact on one or more areas of AI, with exceptionally strong evaluation, reproducibility, and resources, and no unaddressed ethical considerations.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

N/A

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Official Review of Submission29259 by Reviewer swx9

Official Review by Reviewer swx9 📅 24 Jun 2026, 22:05 (modified: 23 Jul 2026, 09:40)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer swx9

📄 Revisions (/revisions?id=riT6ASmv1P)

Summary:

Per my understanding, MIBE has two major contributions:

Dataset/Benchmark, MIB - It's a dataset for measuring multi-subject/object interactions. The training set (silver set, which is automatically annotated) is used for training their metric (MIE). The gold set is the evaluation set which has been labeled by human.

Evaluator, MIE - It is an evaluation metric which has been trained on the silver set. The criteria for evaluation is threefold: Existence (is everyone present?), Appearance (Do the subjects look correct?), Interaction (are relations and actions correct?)

The paper(appendix) mentions the reason why this benchmark is important: they claim that no recent works in multi-concept/subject blending "provides human pairwise gold evaluation labels". Additionally, they claim that recent benchmark do not isolate specific failure types (interactions, occlusion, or identity mix-ups).

For the proposed metric, the authors have mentioned that current metrics do not align with human judgements. They provide examples of how PickScore, ImageReward and HPS are not great for evaluating this usecase.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths:

- The paper explores and expands on a much needed research direction, i.e. Multi-Human/Object composition. There are relatively few benchmarks in this space.
- In terms of usability, given a perfect benchmark/metric, the work is incredibly significant in the industry.
- In this dataset, human interactions are modeled very well. This is one of the major advantages over recent datasets such as MultiHuman-Testbench[1].
- Most of the paper is clearly written. Experiments are easy to follow. Tables and Charts are self explanatory.
- The data and code is made accessible.

Weaknesses:

- The introduction motivates the need for a metric over previous work. The authors have added the motivation for a new benchmark in the supplementary material, but this needs to be included in the introduction to improve readability of the paper.
- The distribution of data is not well presented. Section 3 mentioned the number of pairs and number of references. However, the authors have not provided a thorough dataset distribution of age, ethnicity, gender, hair type, etc. for people. The authors are also urged to provide a distribution of the objects.
- Per my understanding, the subject references have been generated by GPT-Image. This is not ideal, as real faces are harder to generate than synthetic faces.
- To motivate their proposed metrics, the authors mentioned the binding problem in similarity based metrics such as CLIP, Dino, Arcface. However, they have not compared with metrics such as Hungarian ID similarity proposed by MHTB[1]. This metric penalizes heavily for missed/mixed identities.
- Some details on architecture are missing in the paper (see questions).
- A major issue with current multi-human/object generators is that they mix identities of people of the same ethnic background, or similar faces. Is this aspect measured by MIBE?
- The paper missed Withanyone[2] dataset.
- There are no experiments on other multi-subject datasets/protocols. The authors are suggested to perform an analysis showing how dataset A fails to capture certain capabilities of multi-concept gen in an image generation model which have been captured by MIB. (for instance, XVerseBench rates flux2 as A, B, C. MIBE rates it as D, E, F. What is D, E, F telling me that A, B, C does not?)

[1] <https://arxiv.org/abs/2506.20879> (<https://arxiv.org/abs/2506.20879>) [2] <https://arxiv.org/abs/2510.14975> (<https://arxiv.org/abs/2510.14975>)

Quality: 2: not good

Clarity: 3: good

Significance: 4: excellent

Originality: 3: good

Questions:

- As the prompts contain identifiers for each person, "middle eastern man", "south asian woman", is the benchmark measuring a case where two middle eastern men are in the same frame?
- What is the prompt the authors input to the MIE model along with images?
- What are the ranking and diagnostic heads comprising of, and where are they in the model?
- Could the authors clarify theoretically or with experiments, why MIE is better than Hungarian ID similarity, Count Accuracy, Action scores measured by MultiHuman-Testbench[1]
- Could the authors share the data distribution?
- What was the reasoning behind using synthetic subjects?

Limitations:

The authors have discussed limitations, which is well appreciated. Please also discuss and societal impacts.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 5: You are absolutely certain about your assessment. You are very familiar with the related work and checked the math/other details carefully.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

The paper contains references with inaccurate names. Here are some examples:

[1] Shubhankar Borse, **Sungrae Choi, Sanghyun Park, Jihoon Kim, Sayak Kadambi**, Risheek Garrepalli, **Suyeon Yun**, Munawar Hayat, and Fatih Porikli. Multihuman-testbench: Benchmarking image generation for multiple humans. In Advances in Neural Information Processing Systems (NeurIPS), 2025.

[2] **Hao Chen** et al. Xverse: Consistent multi-subject control of identity and semantic attributes via dit modulation. arXiv preprint arXiv:2506.21416, 2025.

[3] Kaiyi Huang, **Kaiqiang Sun, JianENZweiler**, et al. T2i-compbench: A comprehensive benchmark for open-world compositional text-to-image generation. In Advances in Neural Information Processing Systems, 2023.

As such, these papers have the **correct** DOI, links, and title. The authors are however urged to correct the names.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Official Review of Submission29259 by Reviewer yNv5

Official Review by Reviewer yNv5 📅 24 Jun 2026, 11:33 (modified: 23 Jul 2026, 09:40)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yNv5

📄 Revisions (/revisions?id=R4mkoiOmo6)

Summary:

This paper targets the evaluation problem in multi-subject personalized image generation. When several reference identities (2 to 8) and their specified interactions must be rendered, current models frequently omit subjects, lose reference appearance, or misattribute interactions; meanwhile existing metrics (CLIP/DINO similarity, PickScore/ImageReward/HPS, etc.) degrade toward random agreement with human preference as subject count grows

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

strength

1. That standard metrics collapse toward random agreement as subject count grows (Fig. 1) is a genuine, consequential problem. Multi-subject personalized generation is an active area, so a reliable evaluator has clear value.
2. Factorizing "binding" into Existence / Appearance / Interaction is intuitive and gives model developers interpretable, actionable failure attribution rather than a single opaque score.
3. Pairing scalable silver supervision with human pairwise gold labels — and using the gold labels to test whether automatic metrics track humans — is the right way to validate a new metric

weakness

1. The evaluator and its "teacher" share a source. The Silver Set (training labels) is annotated by Gemini-2.5/3.1. MIE is trained on those labels. The paper repeatedly claims MIE far exceeds a "VLM-based scorer (Qwen-7B)" baseline.but that comparison is unfair.
2. The "ground truth" is itself VLM-generated imagery. Reference subjects are GPT-Image-1 generations, and all candidates are generated images. When evaluating identity preservation, the reference identity is synthetic and may not be self-consistent, injecting hard-to-bound noise into the Appearance dimension.
3. A single training matchup in the Silver Set. All 60K training pairs are Nano Banana vs. MOSAIC. MIE scores 0.982 on seen (= these two) but drops ~0.10 to 0.884 on unseen. With training this narrow, the evidence that unseen generalization extends to a broader range of generator families is limited.

Quality: 2: not good

Clarity: 2: not good

Significance: 3: good

Originality: 2: not good

Questions:

See the limitations noted in the Weaknesses section above.

Limitations:

See the limitations noted in the Weaknesses section above.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

None

Code Of Conduct Acknowledgement: Yes
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